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Hearing Difficulties and Depressive Symptoms: The Role of Age, Socioeconomic Status, and Smoking Habits

Introduction

The ability to hear plays a crucial role in social engagement and communication; therefore, its loss can have severe consequences for individuals who experience it. Also, depression represents a major public health concern, and recent evidence suggests that sensory limitations may contribute to depressive symptoms. This case study examines the association between hearing difficulties and depressive symptoms, while also accounting for key confounding variables such as age, household income, and smoking history.

Research Question

Is hearing difficulty associated with depressive symptoms in adults, and how is this relationship influenced by age, household income, and smoking history?

Methodology

Participants

Adult participants from NHANES with complete data on hearing, depression, demographic, economic and drug habits variables were included, representing a diverse U.S. population.

Data Collection

Extracted from NHANES official page.

Questionnaires:

- Hearing (AUQ)
- Depression (DPQ)
- Demographics (DEMO)

Variables

- Some depressive symptoms measured
- Reported hearing difficulty
- Covariates: age, household income, and smoking-related variables, depending on the model

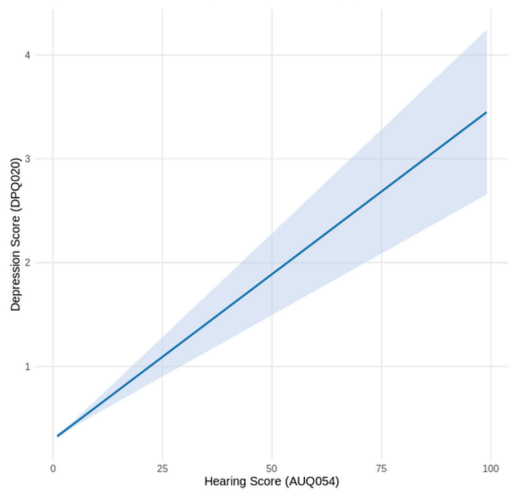
Analysis Pipeline

1. Data cleaning & recoding
2. Exploratory visualizations
3. Simple linear regression
4. Multiple linear regression models:
   - Model 1: Hearing + Age
   - Model 2: Hearing + Household Income
   - Model 3: Hearing + Age started smoking

Results & Discussion

Hearing Loss and Depression

The simple linear regression revealed a **positive** and statistically **significant association** between hearing difficulties and depression . Adults reporting worse hearing abilities tended to show higher levels of depressive symptoms. The regression line indicates a clear **upward trend**, although substantial variability around the line suggests that hearing alone does not fully explain depression outcomes.

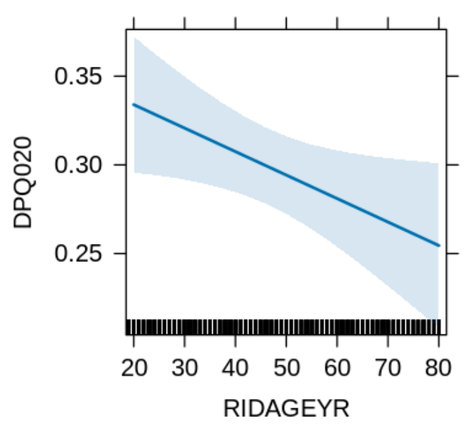


These results support the initial hypothesis that **hearing problems are associated with poorer mental health**. However, as this model does not account for confounding factors, this model serves primarily as an exploratory baseline motivating further adjusted analyses.

Model 1 – Accounting for Age

After adjusting for age, hearing abilities remained **significantly associated** with depression scores, confirming age as an important confounding variable. Compared to the simple linear regression, the effect of hearing was slightly attenuated but remained statistically significant.

The age effect plot shows a modest decrease in predicted depression scores with increasing age, indicating that age alone is not associated with higher depression levels in this adjusted model.

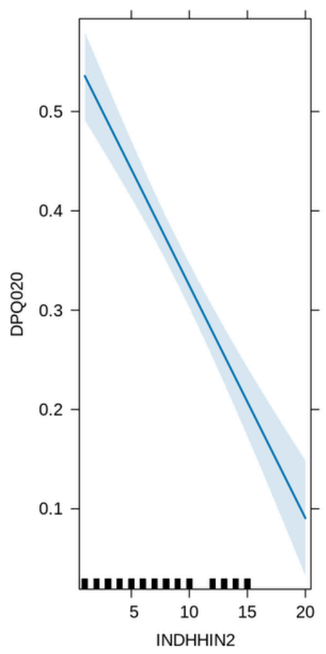


Thus, hearing difficulties contribute independently to depressive symptoms, beyond general age-related patterns in mental health. Model 1 therefore provides more robust evidence for an association between hearing loss and depression.

Model 2 – Accounting for Household Income

There is a clear socioeconomic gradient in depression when household income is included in a regression model. According to the income effect plot, those in lower income categories have the highest projected depression ratings, which gradually decline as income rises. This trend highlights the **significance** of socioeconomic factors in mental health and depressed symptoms.

Thus, Model 2 shows that differences in depression scores are influenced by both hearing health and socioeconomic level. The multidimensional nature of mental health outcomes is further supported by the persistence of the hearing–depression association following income adjustment, which implies that hearing impairments are connected to depressive symptoms beyond poverty-related differences.

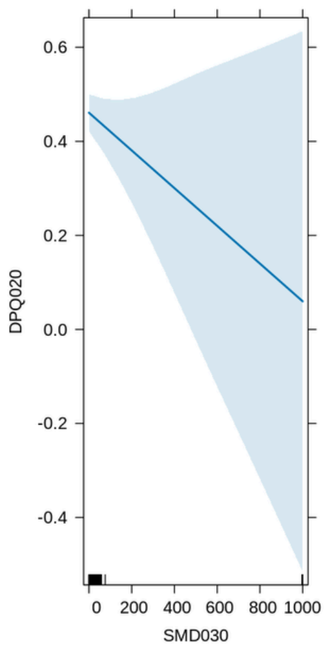


Group Comparisons (T-test)

There is a distinct relationship in depression outcomes when age at smoking initiation is taken into account. Those who **started smoking later are linked to reduced levels of depressed symptoms**, whereas those who started earlier have higher projected depression scores. Fewer observations in these ranges are reflected in increased uncertainty at older beginning ages.

Same for other models, even when smoking habits is taken into consideration, hearing impairments are still highly and favorably linked to depression. The hearing effect plot's continuous rising trend suggests that higher predicted depression scores are correlated with increased hearing impairment.

Model 3 demonstrates that depression is influenced by both behavioral factors and sensory health.



Conclusions

Our hypothesis is **largely** supported by the data:

- Hearing ability showed a **consistent association** with depression, with poorer hearing linked to higher depression scores individually and across all models.
- Age acted as a confounding factor, linking some association between hearing difficulty and depressive symptoms when included in the analysis.
- Household income was inversely related to depression, indicating that socioeconomic status modifies mental health outcomes in adults.
- Smoking history showed a long-term effect, with earlier age of smoking initiation associated with higher predicted depression scores later in life.

Overall, the association between hearing difficulty and depression is shaped by age, socioeconomic status, and smoking history, indicating a multifactorial pathway influencing adult mental health.